

 Google Developer Group • Code for Communities (2nd Edition)

JanSetu AI

AI-Powered Digital Public Infrastructure for Smarter Governance

"From Citizen Voices to Smarter Development Decisions"

 DPI & Governance

 Google Gemini AI

 Multilingual India

Team Details & Track Identification

Team Details

- ⚡ **Project Name:** JanSetu AI
- ⚡ **Team Leader:** Tech Lead / Representative
- ⚡ **Member 1:** AI / ML Specialist
- ⚡ **Member 2:** Full Stack Developer
- ⚡ **Member 3:** Data & Cloud Engineer

Track & Focus

Problem Statement 01:

AI for Digital Public Infrastructure & Governance

Theme: Innovation & Public Good

📍 Addressing core governance challenges across urban & rural India through multimodal AI intelligence.

The Problem: Scattered Citizen Feedback



Fragmented Inputs

Citizen requests (roads, water, healthcare, waste) are scattered across fragmented channels, physical forms, and localized dialects.



Language Barriers

Difficulty in processing voice and text inputs across India's 22+ official languages and local vernacular accents.



Siloed Data

Citizen grievances are processed as isolated complaints rather than combined with demographic & infrastructure indices.



Key Governance Challenge

Policymakers lack an intelligent, automated mechanism to identify high-demand development hotspots and prioritize budget allocation effectively.

JanSetu AI Platform

Unified AI Governance Infrastructure

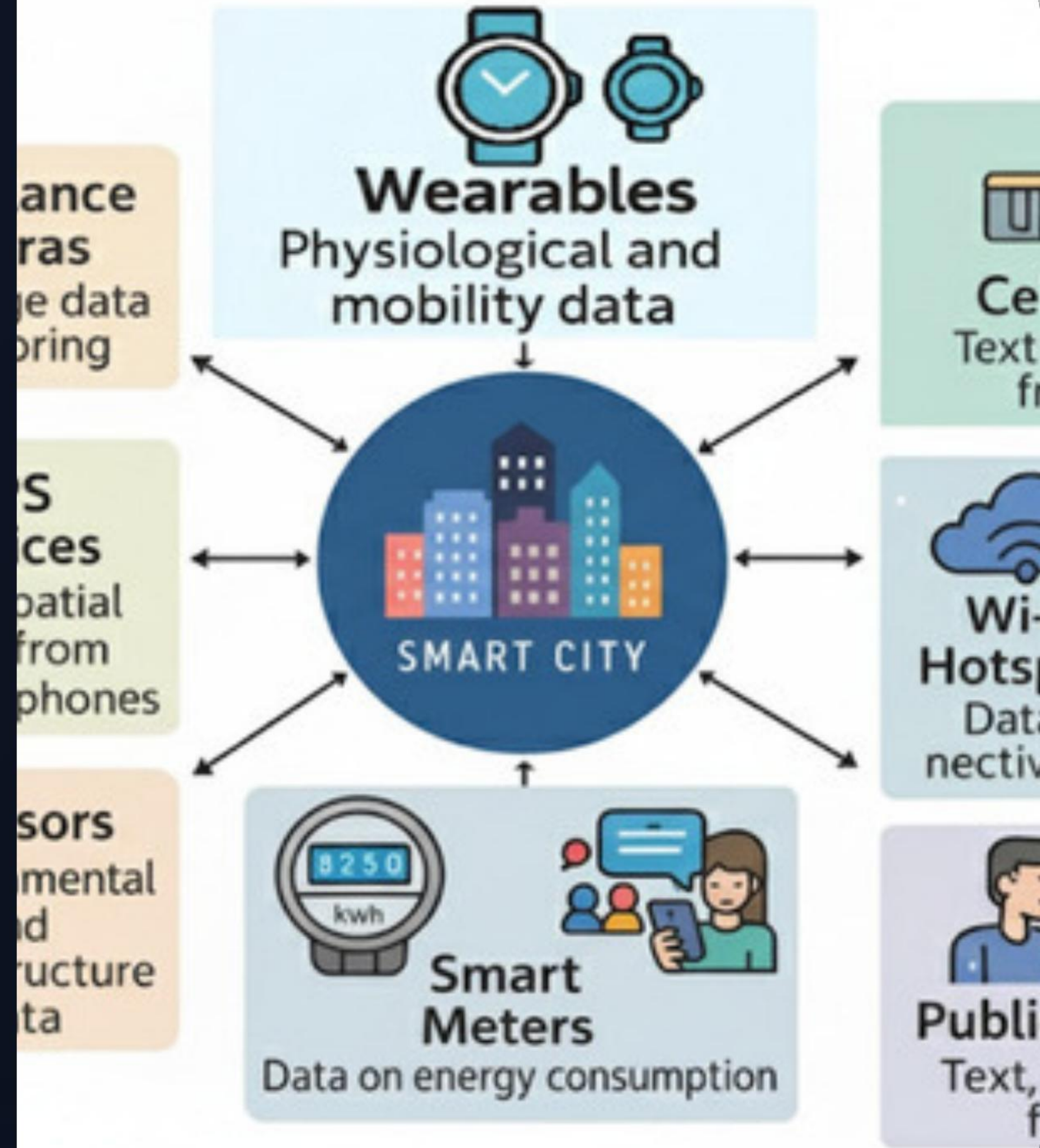


JanSetu AI is a scalable, multilingual **Digital Public Good** that captures citizen development requests via **Voice + Text + WhatsApp/Messaging**.

Multimodal Fusion Engine

Combines real-time citizen demand with demographic density, existing infrastructure gaps, and public investment plans.

- ⚡ Classifies civic needs automatically
- ⚡ Maps geographical demand hotspots
- ⚡ Generates priority score for project recommendations



Innovation & Unique Selling Proposition ⚡

✗ Traditional Systems

- Focus purely on reactive complaint logging.
- Treat issues in isolation (1 broken pipe = 1 ticket).
- Ask only: *"What problem was reported?"*

✓ JanSetu AI Paradigm

- Converts scattered feedback into **Development Intelligence**.
- Aggregates demand into actionable geographical hotspots.
- Answers: **"Where is demand highest, why, & what project to prioritize?"**

Citizen Feedback + **Demographics** + **Infrastructure Data** → **AI Fusion Engine** → **Hotspot Priority Score**

Core Platform Capabilities



1. Multilingual Support

Supports Indian vernacular text & localized speech seamlessly.



2. Voice Reporting

Voice-first entry for digitally under-served rural citizens.



3. Messaging Integration

WhatsApp & Telegram bots for effortless civic reporting.



4. Intent & Urgency AI

Extracts intent, urgency level, and precise issue categories.



5. Hotspot Detection

Clusters high-density demand areas on interactive heatmaps.



6. Demographic Fusion

Analyzes population vulnerability and regional density data.



7. Infrastructure Gap

Compares demand vs existing civic facilities in real-time.



8. Priority Score

AI recommendations tailored for municipal policymakers.

 Google Tech Core Integration

Google Gemini at the Core

Google Gemini Multimodal API powers speech comprehension, multilingual translation, and structured entity extraction.

Speech & Language Layer

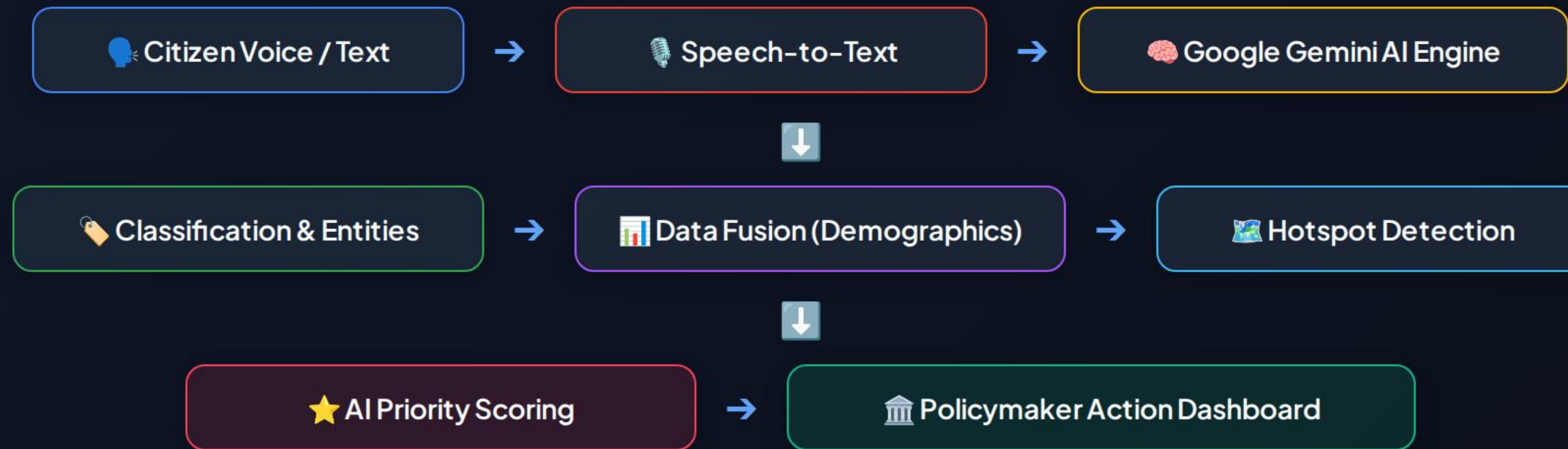
Converts raw regional voice notes to structured text with dialect-resilient NLP.

Entity & Location Extraction

Identifies exact locations, issue types (e.g., healthcare, sanitation), and urgency tags.



System Workflow & Architecture



 **Real-World Example:** Citizen says *"Our ward needs a primary health clinic urgently."* → Gemini extracts location + issue type → Cross-checks population density & distance to nearest clinic → Flags as **High Priority Recommendation (#1)** on Municipal Dashboard.

Technology Stack & Feasibility

Layer	Technologies Used	Role in Hackathon MVP
Frontend	React / Next.js, Tailwind CSS	Interactive Policymaker Dashboard & Citizen Reporting Portal
Backend & API	Node.js / Python, RESTful Services	Data ingestion pipeline & hotspot scoring microservices
AI Core	Google Gemini 1.5 Pro / Vertex AI	Multilingual NLP, voice transcription, entity & intent extraction
Database	Firebase / Firestore / PostgreSQL	Real-time complaint tracking & demographic dataset sync
Mapping	Google Maps Platform (Heatmaps)	Geographical demand visualization and ward-level density rendering

 Hackathon MVP Ready: Fully functional prototype mapping voice input to priority insights.

100% Feasible Concept

Scalability Roadmap & Impact 🚀



Phase 1: Local Pilot

Deploy pilot in **Indore Municipal Corporation** for civic infrastructure demand tracking.



Phase 2: Multi-City

Expand across Smart Cities in Madhya Pradesh & urban centers across India.



Phase 3: National DPI

Build an interoperable national development intelligence layer for state & central ministries.

★ Key Measurable Outcomes

🎯 **Optimized Funds:** Direct public investment to highest-demand zones.

🗣️ **Inclusive Reach:** Voice access for non-literate & rural citizens.

⚡ **Faster Decisioning:** Automated priority matrix for city officials.

Future Vision

Predictive Governance & Global Scale 🌐



Predictive Governance

Forecast emerging infrastructure needs before critical failures occur.



Citizen Feedback Loop

Enable citizens to track real-time project progress and status updates.



BRICS Applicability

Adapt DPI framework for developing nations with similar governance scale.

“JanSetu AI transforms citizen voices into actionable development intelligence, enabling governments to build what communities need most.”

Image Sources



https://www.mdpi.com/smartcities/smartcities-09-00003/article_deploy/html/images/smartcities-09-00003-g002-550.jpg

Source: www.mdpi.com



<https://media.easy-peasy.ai/f38bfdb0-1b75-4c4e-b533-41dc0289c787/b1a8bf81-537a-4c27-a49e-9f810bfbabea.png>

Source: [easy-peasy.ai](https://media.easy-peasy.ai)